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REQUIREMENTS MANAGEMENT PLAN
LITPATH AI: SMART PATHFINDER FOR THESES AND DISSERTATIONS

ASIA PACIFIC COLLEGE
3 HUMABON, MAGALLANES
MAKATI CITY, METRO MANILA 1232

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Introduction

The Requirements Management Plan is a necessary tool for establishing how requirements will be collected, analyzed, documented, and managed throughout the lifecycle of a project. Depending on the type of project there may be both project and product requirements. It is easy to unintentionally omit requirements, fail to document them, or leave requirements incomplete without a tool to properly manage them.

The Requirements Management Plan establishes how requirements will be collected, analyzed, documented, and managed throughout the lifecycle of the LitPath AI project. LitPath AI is a standalone AI-powered web application that provides advanced, AI-driven academic thesis and dissertation search capabilities for the DOST-STII Library. While it operates independently as its own platform, it relies on the DOST-STII Library's existing thesis and dissertation database as its primary data source, maintaining a read-only connection to ensure no modifications are made to library records.

Requirements are divided into two categories: project requirements (non-technical requirements related to delivery, integration, and compliance) and product requirements (technical specifications for system functionality, performance, and quality). Inputs to this plan include the LitPath AI Project Charter, SRS document, the Stakeholder Management Strategy, and the Stakeholder Analysis Matrix. The Project Sponsor is Khasian Eunice M. Romulo, Science Research Specialist II / Unit Head of the Digital Information and Processing Delivery Unit, DOST-STII. The Project Manager is Tracie Tomon (HecTech Group Leader), authorized to negotiate resources, delegate responsibilities, and ensure timely completion.

Requirements Management Approach

The requirements management approach is the methodology the project team will use to identify, analyze, document, and manage the project's requirements. Many organizations use a standard approach for all projects, but based on the characteristics of each project, this approach may require some changes. The PMBOK defines this approach as "How requirements activities will be planned, tracked, and reported."

The requirements management approach is the methodology the project team will use to identify, analyze, document, and manage the project's requirements. For LitPath AI, this approach is broken down into four areas: requirements identification, requirements analysis, requirements documentation, and ongoing requirements management.

Requirements Identification: The LitPath AI team collected requirements through analysis of the DOST-STII OPAC system's limitations, benchmarking tools like UP Diliman, Tuklas and Google Scholar, and consultations with the project client and faculty advisers. Methods included document reviews, client meetings, and iterative SRS development to ensure all requirements were captured.

Requirements Analysis: The team categorized requirements into functional and non-functional groups and evaluated each based on priority, feasibility, and technical constraints. A Work Breakdown Structure (WBS) was used to organize these requirements into manageable components and map them to corresponding deliverables. Traceability was established through use cases and business rules, while accountability, metrics, and acceptance criteria were defined to ensure clear completion standards.

Requirements Documentation: All identified and analyzed requirements were documented with unique IDs for functional requirements and supporting use cases and business rules. These were integrated into the project plan and tracked to ensure proper implementation and reporting.

Ongoing Requirements Management: Throughout the project lifecycle, the Project Manager ensures continuous monitoring of requirement status and addresses any issues raised by the team. Any changes follow the established change control process, and final approval of all requirements is secured from the Project Sponsor before project closure.

Configuration Management

In order to effectively manage a project, communication must be managed and controlled. Additionally, every effort must be taken to identify requirements thoroughly during project initiation and planning. However, there are often situations which require changes to a project or its requirements. In these situations, it is important to utilize configuration management to consider proposed changes, establish a process to review and approve any proposed changes, and to implement and communicate these changes to the stakeholders. As stated in the PMBOK: "Configuration management activities such as how changes to the product, service or result requirements will be initiated, how impacts will be analyzed, how they will be traced, tracked, and reported, as well as the authorization levels required to approve these changes."

To effectively manage the LitPath AI project, communication must be managed and controlled. While every effort has been made to identify requirements thoroughly during project initiation and planning, situations may arise that require changes. In these situations, it is important to utilize configuration management to consider proposed changes, establish a review and approval process, implement approved changes, and communicate those changes to all stakeholders.

Documentation and Version Control: All project documentation is version-controlled and tracked through the SRS revision history. The LitPath AI SRS has been through 15 documented revision cycles (Draft v1.0 through Draft v3.0), with each revision attributed to specific team members and dated. Any proposed changes to project requirements must be reviewed by the HecTech team and have written approval from the Project Sponsor, Ms. Khasian Eunice Romulo, before any documentation changes are made. Once approved, the Project Manager is responsible for communicating the change to all project stakeholders.

Change Control: Any proposed changes to project requirements must be carefully considered before approval and implementation, as such changes are likely to impact project scope, time, and cost. Any change that affects milestone dates must be approved by the Project Sponsor. Changes affecting the read-only connection to the DOST-STII database or the Gemini AI service integration must also be reviewed with the alternate POC, Mr. Jonathan Abalon, who is specifically responsible for system access and technical support and has identified system integration risks as a primary concern.

REQUIREMENTS PRIORITIZATION PROCESS

Prioritizing requirements is an important part of requirements management. Developers or service providers do not always know what requirements are most important to a customer. Conversely, customers do not always understand the scope, time, and cost impacts of their requirements on a project. Collaboration among all stakeholders is a necessary part of establishing project requirement priorities. If cuts need to be made to scope, time, or cost, this list of priorities will provide a better understanding of where a project can or cannot focus on dealing with the constraints placed upon it. One way to do this is to group requirements into priority categories such as high, medium, and low priority based upon the importance of the requirement. There may be hundreds of requirements in a large project so this type of categorically based method would be helpful. NOTE: there are many methods by which priorities are determined, and these should be explored based on the size and complexity of the project.

Prioritizing requirements is an important part of requirements management for LitPath AI, as the project operates under a fixed deployment deadline of June–July 2026 and limited development resources. The HecTech project team, in coordination with key stakeholders, established priorities for all project requirements using a three-level scale.

Requirement priorities were established in coordination with the key stakeholders and managed closely per the Stakeholder Management Strategy: Ms. Khasian Eunice Romulo and Mr. Jonathan

Abalon (High Power, High Interest). Their primary concerns are search relevance improvement and system integration stability, which directly informed the High priority classification of core functional requirements. The chart below illustrates the three priority levels and defines how requirements are grouped:

Priority Level	Definition
High	Must be delivered for the system to function which is directly tied to core client needs and project charter deliverables.
Medium	Important features that significantly improve usability but are not blockers for deployment
Low	Additional features that can be implemented later if necessary

As the project moves forward, and additional constraints are identified, the project team and stakeholders may meet to determine which requirements must be achieved, which can be re-baselined, or which can be omitted. These determinations will be made collaboratively based on the priority levels above. Any approved changes will be reflected in updated project documentation and communicated to all stakeholders.

Product Metrics

Product metrics are an important part of determining a project's success. There must be some quantitative characteristics to measure against in order to gauge the progress and success of the project. Product metrics are usually technical in nature though not always. Such metrics may consist of performance, quality, or cost specifications. Metrics may also be based on the product requirements identified for a given project.

Product metrics are an important part of determining the success of LitPath AI. The following metrics are drawn from the LitPath AI Project Charter and SRS and represent the quantitative benchmarks the system must meet or exceed.

Search Performance:

- Search response time must be up to the industry standard of under 30 seconds, improved from the current 2–3 minutes in the existing OPAC system
- Search relevance accuracy must reach at least 70% relevance in the top 10 results
- Time spent by users filtering irrelevant results must be reduced from approximately 15 minutes to just a few minutes

User Satisfaction:

- The system must achieve a user satisfaction score of at least 3.5 out of 5 for search feature usability, to be confirmed through usability testing during the Bug Fixes and Usability Testing phase (May–June 2026)

Functional Coverage:
- Citation formats supported: APA 7th Edition, MLA 9th Edition, Chicago Manual of Style, and IEEE (4 formats)
- All functional requirements must be delivered and verified before final deployment

System Reliability and Security:
- Each authenticated user may have only one active session at a time
- Passwords must be stored as hashed values and all sessions must be managed with secure tokens

Requirements Traceability Matrix

The requirements traceability matrix is a tool to ensure that deliverables meet the requirements of the project. The matrix provides a thread from the established and agreed upon project requirements, through the project's various phases, and through to completion/implementation. This ensures that the product specifications and features satisfy the requirements on which they were based. Any interim project tasks associated with the requirements should be included. An example of this are test cases which will utilize metrics, based on the product requirements, which are linked to the project charter and design document. This allows a team member or stakeholder to follow a requirement through all tasks required for its completion.

The requirements traceability matrix ensures that all LitPath AI deliverables meet the requirements established in the Project Charter and SRS v3.0. It provides a thread from each requirement through its corresponding use case, charter reference, and priority level, ensuring that all features can be traced from inception through to testing and deployment.

Project Name	LitPathAI	**Project Manager**	Tracie Tomon	
Req #	**UC Ref**	**Requirement Description**	**Test Case Ref**	**Test Cases**

F10	UC-01	The system shall support user authentication with email and password through login.	TC_MA	TC_MA_001 to TC_MA_030
F11	UC-01	The system shall manage user sessions with secure tokens. Each authenticated user can have only one active session at a time.	TC_MA	TC_MA_008, TC_MA_039
F12	UC-01	The system shall allow authenticated users to change their passwords.	TC_MA	TC_MA_031 to TC_MA_050
F13	UC-01	The system shall allow authenticated users to deactivate their accounts.	TC_MA	TC_MA_049
F15	UC-01	The system shall support multiple user roles: User and Admin with appropriate access permissions.	TC_MA	TC_MA_006, TC_MA_007
F1	UC-02	The system shall allow users to enter natural language research queries in a search input field.	TC_ST/D	TC_ST/D_001 to TC_ST/D_010
F2	UC-02	The system shall perform	TC_ST/D	TC_ST/D_006 to TC_ST/D_025

		semantic search across indexed thesis documents using vector embeddings and return relevant results ranked by similarity.		
F4	UC-02	The system shall display thesis metadata for each search result including title, author, publication year, abstract, degree, subject/s, and university.	TC_ST/D	TC_ST/D_026 to TC_ST/D_050
F4	UC-03	The system shall display thesis metadata for each search result including title, author, publication year, abstract, degree, subject/s, and university.	TC_VTD	TC_VTD_001 to TC_VTD_021
F5	UC-04	The system shall allow users to bookmark/save thesis documents for later reference.	TC_BMK	TC_BMK_001 to TC_BMK_013
F6	UC-04	The system shall persist bookmarks for authenticated users in the database.	TC_BMK	TC_BMK_007 to TC_BMK_010

F7	UC-05	The system shall generate properly formatted citations in APA, MLA, Chicago, and IEEE citation styles.	TC_CTD	TC_CTD_001 to TC_CTD_012
F8	UC-06	The system shall maintain research history tracking all user search sessions with timestamps.	TC_VSH	TC_VSH_001 to TC_VSH_010
F9	UC-06	The system shall allow users to reload and revisit previous research sessions from history.	TC_VSH	TC_VSH_011 to TC_VSH_020
F14	UC-07.1	The system shall allow users to provide feedback on search results including relevance ratings (star ratings) and comments.	TC_EF	TC_EF_001 to TC_EF_050
F17	UC-07.2	The system shall allow admins to respond to feedback and generate reports for usage analytics.	TC_MF	TC_MF_001 to TC_MF_075
F17	UC-08	The system shall allow admins to respond to feedback and	TC_UA	TC_UA_001 to TC_UA_085

		generate reports for usage analytics.		
F3	UC-09	The system shall generate AI-powered overview summaries synthesizing information from retrieved thesis documents with inline reference citations.	TC_GAA	TC_GAA_001 to TC_GAA_025
F16	UC-09	The system shall extract and index metadata from thesis documents including title, author, year, abstract, degree, and subjects.	TC_GAA	TC_GAA_026 to TC_GAA_050

SPONSOR ACCEPTANCE

Approved by the Project Sponsor:

____ Date: April 30, 2026

Khasian Eunice Romulo
Science Research Specialist II / Unit Head
(Digital Information and Processing Delivery Unit)